

Job Market Pulse

Real-Time Skill Demand and Salary Intelligence Report

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1. Project Overview

Job Market Pulse is an end-to-end data analytics project that analyses 90,000 real-world global AI and data science job records to provide actionable salary intelligence. The project answers key questions for job seekers: which skills pay more, how remote work affects salary, and which industries and countries offer the best compensation.

Dataset Summary

Attribute	Details
Source	Global AI and Data Jobs Dataset (Kaggle)
Total Records	90,000
Total Features	35 columns
Years Covered	2020 - 2026
Null Values	Zero - completely clean
Salary Column	salary_in_usd

2. Key Insights

Insight 1 - Skill Category Salary Lift

Machine Learning and AI professionals earn 5% above market average of \$96,546, while Data Analytics professionals earn 28% below market average.

Skill Category	Salary Lift	Avg Salary USD
Machine Learning / AI	+5.0%	~\$101,373
Data Science	-2.6%	~\$94,028
Data Analytics	-28.1%	~\$69,416

Key Takeaway: Transitioning from Data Analytics to Machine Learning/AI increases salary by 33% - the single most impactful career move in the data field.

Insight 2 - Top Countries by Average Salary

USA leads globally with average salaries of approximately \$125,000, followed by Singapore and Australia.

Rank	Country	Avg Salary USD
1	USA	~\$125,000
2	Singapore	~\$115,000
3	Australia	~\$108,000
4	Canada	~\$107,000
5	UK	~\$103,000
6	Netherlands	~\$103,000
7	Germany	~\$98,000
8	Japan	~\$97,000
9	France	~\$96,000
10	UAE	~\$95,000

Key Takeaway: USA pays 30% more than UAE for same data roles - making it the most lucrative market globally.

Insight 3 - Top AI Specializations

Generative AI and LLM specializations command the highest salaries in 2025-2026.

Rank	AI Specialization	Avg Salary USD
1	Generative AI	~\$98,000
2	LLM	~\$97,500
3	Analytics	~\$97,000
4	MLOps	~\$97,000
5	Computer Vision	~\$96,500

Key Takeaway: Learning Generative AI and LLM skills is the highest ROI investment for data professionals in 2025-2026.

Insight 4 - Salary Trend 2020-2026

Data science salaries peaked in 2021, dropped in 2022, hit lowest in 2024, then recovered strongly in 2025-2026.

Key Takeaway: 2025-2026 is a good time to negotiate salary increases as market recovers.

Insight 5 - Top Industries

Retail and Gaming industries pay more than Finance and Tech for data roles.

Key Takeaway: Don't limit job search to Tech - Retail and Gaming offer competitive salaries.

3. Salary Prediction Model

Metric	Value
Algorithm	Random Forest Regression
Training Records	72,000
Testing Records	18,000
R2 Score	0.54
Mean Absolute Error	\$21,769
Sample Prediction	Senior + Remote + Large + ML/AI + 2025 = \$107,138

4. Tech Stack

Technology	Purpose
Python	Data analysis and ML
Pandas and NumPy	Data cleaning and manipulation
Matplotlib and Seaborn	Data visualisation
Scikit-learn	Random Forest model
Streamlit	Interactive dashboard
Streamlit Cloud	Free deployment
GitHub	Version control

5. Project Links

Live Dashboard: <https://job-market-pulse-hqc6evyqecnn3fvhdbunup.streamlit.app/>

GitHub Repository: <https://github.com/Althaf-725/job-market-pulse>

Portfolio: <https://althaf-data-analyst-portfolio-rdx1.vercel.app/>

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